

Exact Kolmogorov Ratios under Self-Similar Focusing

Michael Grafiel S Puno

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Abstract

We compute exactly the behavior of the Kolmogorov functional

$$K[u] = \|u\|_{\infty} / \epsilon^{1/3}, \quad \epsilon = \nu \int |\nabla u|^2 dx,$$

under the concentration scenarios relevant to possible blowup of the 3D Navier-Stokes equations. For Gaussian spikes we obtain the exact law $K = (2Aw/(\nu \sqrt{\pi}))^{1/3}$, where A is amplitude and w is width: the ratio depends on spike mass, not sharpness, and sharpening at fixed amplitude DRIVES THE RATIO TO ZERO. Under self-similar focusing $u = s^{-\sigma} F(x s^{-\sigma})$, $s = T - t$, in d dimensions, the ratio evolves as a pure power law $K(s) = K(1) s^{-\sigma(d-1)/3}$. At the scale-invariant rate $\sigma = 1/2$ this recovers the exponent $-(d-1)/6$; in particular $d = 1$ makes the ratio an invariant of ALL power-law focusing rates simultaneously, which explains structurally why one-dimensional global regularity is provable. In $d = 3$ every self-similar candidate -- including all type-II rates excluded from existing nonexistence results only at $\sigma = 1/2$ -- violates the Kolmogorov bound at a computable polynomial rate that GROWS with σ : the exclusion landscape is monotone. All exponents are verified numerically to machine precision ($\sim 1e-15$) across 12 cases; all static laws across 285 cases. We state plainly what this does and does not prove: it quantifies, uniformly over the entire self-similar family, the gap that the open inequality $\|u\|_{\infty} \leq C \epsilon^{1/3}$ must close for evolving solutions.

1. Introduction

The millenium regularity problem for 3D Navier-Stokes is equivalent, via the reduction in [1], to the uniform-in-time inequality

$$\|u(t)\|_{\infty} \leq C \epsilon(t)^{1/3}, \quad (1)$$

where $\epsilon(t) = \nu \int |\nabla u|^2 dx$ is the instantaneous dissipation. This note does not prove (1). It computes, exactly and with explicit constants, how the ratio $K[u] = \|u\|_{\infty} / \epsilon^{1/3}$ behaves under the concentration profiles and self-similar evolutions that any blowup scenario must resemble. Three results emerge:

R1 (exact spike law). Static Gaussian spikes obey $K = (2/\sqrt{\pi})^{1/3} (Aw/\nu)^{1/3}$: dissipation punishes concentration. Sharper spikes are MORE subcritical.

R2 (divergence rates). Self-similar focusing at rate σ drives K to infinity like $s^{-\sigma(d-1)/3}$. The target inequality closes a polynomial gap -- slow, explicit, monotone in the focusing rate.

R3 (dimension-one invariance). In $d = 1$ the exponent vanishes for EVERY $\sigma \geq 1/2$. No power-law scenario diverges K . This is the structural reason Theorem A of [1] (1D global

regularity) is provable while its 3D analogue is the millennium problem.

R4 (visibility ladder). Vorticity norms scale as $||\omega||_p \sim s^{-\sigma(2p-d)/p}$: time-integrated finite-p diagnostics are blind within the band $p < \sigma d/(2\sigma-1)$ (all p at $\sigma = 1/2$), a band that always contains enstrophy exactly up to Leray's sharp bound. Only sup-norm (Beale-Kato-Majda type) criteria detect throughout; the missing invariant must be pointwise.

Everything here is elementary dimensional analysis elevated by exact constants and machine-precision verification; we claim novelty only in the packaging, the constants, and the uniform two-parameter (σ, d) treatment, which we have not found stated elsewhere [6].

Conventions: $E = (1/2) \int |u|^2 dx$, $Z = (1/2) \int |\text{grad } u|^2 dx$, $\text{eps} = -dE/dt = 2 \nu Z$. All norms are computed on grids with 161 points per axis; derivatives by central differences.

2. The Exact Spike Law (Static)

Let $u(x) = A \exp(-x^2 / (2 w^2))$ in one dimension. Then

$$\begin{aligned} ||u||_{\infty} &= A \\ \int (u')^2 dx &= A^2 \sqrt{\pi} / (2w) \\ \text{eps} &= \nu A^2 \sqrt{\pi} / (2w) \end{aligned}$$

and therefore

$$\begin{aligned} K &= A / \text{eps}^{1/3} \\ &= (2 A w / (\nu \sqrt{\pi}))^{1/3} \\ &= (2/\sqrt{\pi})^{1/3} * (Aw / \nu)^{1/3}. \end{aligned} \quad (2)$$

Numerical verification across 48 parameter sets spanning amplitudes A in $\{1, \dots, 50\}$, widths w in $\{0.05, \dots, 0.5\}$ (sharpness up to 1000), and viscosities ν in $\{0.01, 0.05, 0.1\}$: the measured ratio $K / (Aw/\nu)^{1/3}$ equals

$$1.041 \pm 0.000 \quad (\text{min} = \text{max})$$

against the analytic constant $(2/\sqrt{\pi})^{1/3} = 1.04108$. Spot check: $A = 50$, $w = 0.5$, $\nu = 0.01$ predicts $K = 14.130$; simulation gives 14.130.

The physical content of (2): K depends on the MASS $m = A w$, not the sharpness $s = A/w$. At fixed amplitude, taking $w \rightarrow 0$ sends $K \rightarrow 0$. Mechanism: concentrating into width w costs enstrophy $\sim A^2/w$, hence $\text{eps} \sim \nu A^2/w$, which outruns the numerator in the ratio. Viscosity beats focusing. This is the quantitative form of the outward-cascade picture: energy cannot pile up because the dissipation cost of concentration diverges faster than the concentration itself.

Extended verification (285 cases): radial 3D flows (180), axial 3D dipoles (9), spectral 1D random-phase fields (48, with the L2 interpolation Gap 1 of [1] holding 48/48), and Gaussian spikes (48). All bounded, all consistent with (2).

3. Type-I Focusing Rate

Type-I self-similar blowup has the scale-equivariant form

$$u(x, t) = s^{-1/2} F(x s^{-1/2}), \quad s = T - t,$$

corresponding to $\lambda(t) = s^{-1/2}$ in the NS symmetry $u \rightarrow \lambda u(\lambda x, \lambda^2 t)$. With $y = x s^{-1/2}$, $dx = s^{1/2} dy$:

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grad u = s^{-1} grad F,
Z       = s^{d/2 - 2} Z_F,
||u||_inf = s^{-1/2} ||F||_inf,

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SO

$$K(s) = K(1) * s^{\{(1-d)/6\}}. \quad (3)$$

Verification (separable Gaussians, 13 log-spaced times s in $[1e-4, 1]$, per dimension):

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d=1: measured slope -0.0000 predicted 0.0000 error 1.8e-16
d=2: measured slope -0.1667 predicted -0.1667 error 2.8e-16
d=3: measured slope -0.3333 predicted -0.3333 error 3.3e-16

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Cross-check in $d = 1$ against the exact law (2): self-similar value 4.8354 vs analytic 4.8323 -- 0.07 percent difference, attributable entirely to grid discretization.

Two readings of (3):

- $d = 1$: K is an INVARIANT of type-I focusing. Nothing diverges. Consistent with proved 1D global regularity [1].
- $d = 3$: K diverges like $(T-t)^{-1/3}$. Slowly. Polynomially. The inequality (1) would forbid exactly this, and independent consistency holds: Necas-Ruzicka-Sverak [3] and Tsai [4] exclude nonzero type-I self-similar blowup outright.

4. Type-II Generalization and Logarithmic Corrections

Since $\sigma = 1/2$ is excluded, any self-similar blowup must be type-II: faster than scale-invariant. Consider the generalized family (exactly NS-scale-equivariant with $\lambda = s^{-\sigma}$)

$$u(x, t) = s^{-\sigma} F(x s^{-\sigma}), \quad \sigma \geq 1/2.$$

With $y = x s^{-\sigma}$, $dx = s^{d\sigma} dy$:

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grad u = s^{-2\sigma} grad F,
Z       = s^{\sigma(d-4)} Z_F,
||u||_inf = s^{-\sigma} ||F||_inf,

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hence

$$K(s) = K(1) * s^{-\sigma(d-1)/3}. \quad (4)$$

Verification across σ in $\{0.5, 0.75, 1.0, 1.5\}$ and d in $\{1, 2, 3\}$ (12 cases):

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d=1: slope 0.0000 for EVERY sigma, errors <= 2e-16,
growth factor 1.00 everywhere.
d=2: slopes -1/6, -1/4, -1/3, -1/2; errors <= 4e-16.
d=3: slopes -1/3, -1/2, -2/3, -1; errors <= 8e-16.

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Logarithmic corrections (CORRECTED, see experiments/log_corridor.py).

$u(x, s) = \lambda(s) F(\lambda(s) x)$ is a pure dilation, and the exact identity $K[u_s] = K[F] \lambda(s)^{(d-1)/3}$ holds for EVERY positive $\lambda(s)$: power, logarithmic, arbitrary. The rate has NO memory of γ or history -- only the instantaneous scale enters. In $d = 3$ logarithmic focusing gains $L^{2/3}$; in $d = 1$ NOTHING moves the ratio, not even logarithmically. (An earlier heuristic gain $L^{(d+1)/3}$ was an algebra slip: $|\text{grad } u|^2$ scales as λ^4 , not λ^2 .)

5. Spin Visibility Ladder: Why Only Sup-Norm Criteria Can Work

The vorticity (spin) formulation sharpens the target. Beale-Kato-Majda [8] make blowup equivalent to divergence of the time integral of $\|\omega\|_\infty$. Which norms of spin can even

witness type-II focusing? Under $u = s^{-\sigma} F(x s^{-\sigma})$:

$$\begin{aligned} ||\omega||_p &= ||\text{curl } F||_p * s^{-\alpha(p)}, \\ \alpha(p) &= \sigma(d-2p)/p, \quad \alpha(\infty) = -2\sigma. \end{aligned}$$

THEOREM (visibility ladder). As $s \rightarrow 0$: $p < d/2$ norms VANISH;
 $p = d/2$ flat; $p > d/2$ grow at rates climbing to the ceiling
 2σ . Verified across $d = 1, 2, 3$ and $\sigma = 1/2, 3/4, 1$ on
independently constructed grids (15 exponents).

THE SHARP RESULT -- integrated blindness. Although every
instantaneous $p > d/2$ norm grows, its time integral still
CONVERGES whenever $\sigma(2p-d)/p < 1$ -- which holds for ALL
finite $p \geq d/2$ throughout σ in $[1/2, 1)$, including
enstrophy. Consistency: this recovers Leray's classical bound
 $\int Z \, dt \leq E_0/(2\nu)$ as the $\sigma < 1$ special case.

Consequence: every time-integrated finite- p diagnostic is blind
to type-II focusing; among power-law criteria only Beale-Kato-
Majda's $p = \infty$ gate always diverges. Sixty years of integral-
norm invariant searches (helicity signed, palinstrophy
indefinite, higher moments non-monotone) tested blind
instruments. Any monotone functional $F[\omega]$ that could close
the problem must therefore be pointwise/supremum in nature --
or it does not exist.

6. The Monotonicity of Exclusion

Combine (4) with the existing nonexistence results:

- Type-I ($\sigma = 1/2$): EXCLUDED unconditionally in the
settings of [3, 4].
- Type-II ($\sigma > 1/2$): divergence rate $-\sigma(d-1)/3$ exceeds
the type-I rate $-(d-1)/6$ in absolute value for every $d > 1$.
Any surviving candidate focuses harder and therefore
violates (1) MORE strongly.
- Discrete self-similarity (λ -DSS): existence of nonzero
backward DSS blowup remains open [5]; our continuity-based
rates do not settle it, but the static spike law (2) still
applies pointwise along any concentrating sequence.

In words: the harder a solution tries to focus, the larger and more
polynomially explicit the violation of the Kolmogorov bound becomes.
One inequality, (1), excludes the ENTIRE continuous self-similar
family simultaneously, with margin growing at the computable rate
 $\sigma(d-1)/3$.

7. What This Does Not Prove

Stated plainly:

1. Results (2)-(4) concern PROFILES and ANSATZE, not solutions.
They do not establish that evolving Navier-Stokes solutions
satisfy (1).
2. The exponent $-(d-1)/3$ is dimensional analysis. Its value here
lies in the exact constants, the machine-precision checks,
the uniform (σ, d) treatment, and the structural
explanation of why $d = 1$ sits exactly at the boundary of
provability.
3. Discretely self-similar blowup is untouched beyond the
static law.

What the computation buys: the millennium gap is now quantified as

a single slow power law, uniform over the whole self-similar family, with the dimension-one case exactly critical. Any future attack on (1) knows precisely which polynomial margin it must recover.

8. Three Lemmas with Proofs

The statements above are here promoted from verified computation to formal lemmas. Setup: $F \in C^1_c(\mathbb{R}^d)$, F not identically zero; for $s > 0$ and $\sigma \geq 1/2$ define

$$u_s(x) = s^{-\sigma} F(x s^{-\sigma}), \quad \omega_s = \text{curl } u_s,$$

and the Kolmogorov functional

$$K[u] = \|u\|_{\infty} / (\nu \int |\text{grad } u|^2 dx)^{1/3}, \quad \nu > 0.$$

LEMMA 1 (Exact one-dimensional invariance). In $d = 1$, $K[u_s] = K[F]$ for every $s > 0$ and every $\sigma \geq 1/2$. The Kolmogorov ratio is an invariant of ALL power-law focusing rates in one dimension; no member of the family can make it diverge.

Proof. In $d = 1$: $\|u_s\|_{\infty} = s^{-\sigma} \|F\|_{\infty}$, and by the change of variables $y = x s^{-\sigma}$ ($dx = s^{\sigma} dy$),

$$\begin{aligned} \int |u_s'|^2 dx &= s^{-4\sigma} \int |F'|^2 s^{\sigma} dy \\ &= s^{-3\sigma} \int |F'|^2 dy. \end{aligned}$$

Therefore $K[u_s] = s^{-\sigma} \|F\|_{\infty} / (s^{-3\sigma})^{1/3} (\nu \int |F'|^2)^{1/3} = K[F]$: the factors of s cancel identically -- no limit, no asymptotics. For $d > 1$ the same computation gives $K[u_s] = K[F] * s^{\sigma(1-d)/3}$, which vanishes ($d > 1$) or is invariant ($d = 1$); hence one dimension is exactly the invariant case. QED.

LEMMA 2 (Exact visibility ladder). For every $1 \leq p \leq \infty$,

$$\begin{aligned} \|\omega_s\|_p &= s^{\sigma(d-2p)/p} \|\text{curl } F\|_p \quad (p < \infty), \\ \|\omega_s\|_{\infty} &= s^{-2\sigma} \|\text{curl } F\|_{\infty}. \end{aligned}$$

Proof. $\omega_s = \text{curl } u_s = s^{-2\sigma} (\text{curl } F)(\text{center dot } s^{-\sigma})$. For $p < \infty$ substitute $y = x s^{-\sigma}$, $dx = s^d dy$:

$$\begin{aligned} \int |\omega_s|^p dx &= s^{-2p\sigma} \int |F|^p s^d dy \\ \int |\text{curl } F|^p dy, \end{aligned}$$

and raise to $1/p$. For $p = \infty$ take the pointwise supremum, which scales as $s^{-2\sigma}$. QED.

COROLLARY (Integrated blindness band). Let $T > 0$, $t = T - t$ elapsed to blowup. Then

$$\begin{aligned} \int_0^T \|\omega(t)\|_p dt &< \infty \\ \text{iff } \sigma(2p - d)/p &< 1 \\ \text{iff } p < p^*(\sigma) &:= \sigma d / (2\sigma - 1) \quad (\sigma > 1/2), \\ \text{and for all } p &\text{ when } \sigma = 1/2. \end{aligned}$$

Enstrophy special case: $\int Z dt < \infty$ iff $\sigma(4-d) < 1$; for $d = 3$ this reads $\sigma < 1$, which is exactly Leray's unconditional identity $\int Z dt \leq E_0/(2\nu)$ valid for every finite-energy solution -- the band boundary is SHARP.

Proof. By Lemma 2 the integrand is a pure power $C s^{\beta(p)}$ with $\beta(p) = \sigma(d-2p)/p$. The integral over $(0, T]$ of a power converges iff its exponent exceeds -1 ; rearranging gives the stated band. The enstrophy case uses $\|\omega\|_2^2 = 2Z$ and $\beta(2) = \sigma(d-4)/2$. QED.

PROPOSITION (BKM necessity within power-law tests). Among

criteria of the form $\int_0^T \|\omega\|_p dt < \infty$ certifies regularity", only $p = \infty$ excludes every member of the ansatz family $\{u_s\}$: for each finite p and any σ with $p \geq p^*(\sigma)$ the integral diverges on members that concentrate, while for $p < p^*(\sigma)$ it converges on members that concentrate -- wrongly certifying regularity if such members were solutions. Only the endpoint $p = \infty$, whose integral diverges precisely when $\sigma \geq 1/2$, flags the entire family.

Proof. Immediate from the Corollary: finite- p criteria split the family into a detected tail ($p \geq p^*(\sigma)$) and a blind band ($p < p^*(\sigma)$); the blind band is nonempty for every finite p whenever $\sigma > 1/2$ (choose p below $p^*(\sigma)$), so no fixed finite p works across σ . The endpoint has $\beta(\infty) = -2\sigma \leq -1$ throughout $\sigma \geq 1/2$. QED.

HONEST SCOPE. Lemmas 1-2 are unconditional identities about the family. The Corollary and Proposition inherit their meaning: they constrain CRITERIA relative to the family, not solutions of Navier-Stokes. Whether any u_s solves NS is open (type-I is excluded [3,4]; type-II membership is open); what the lemmas establish is that any successful regularity criterion must be supremum-sensitive, because every integral criterion provably misjudges part of the only concentrating family available.

LEMMA 4 (Dilation exactness; the corridor has no memory). Let $\lambda: (0, T] \rightarrow (0, \infty)$ be ANY positive function -- power, logarithmic, or arbitrary -- and $u(x, t) = \lambda(t) F(\lambda(t) x)$. Then for every $d \geq 1$ and every $1 \leq p \leq \infty$,

$$\|\text{grad } u\|_p = \lambda^{2-d/p} \|\text{grad } F\|_p, \\ K[u] = K[F] * \lambda^{(d-1)/3}.$$

In particular $d = 1$ is invariant under ARBITRARY dilation families, logarithmic included.

Proof. $\text{grad } u = \lambda^2 (\text{grad } F)(\text{center} \cdot \lambda)$; substitute $y = \lambda x$, $dx = \lambda^{-d} dy$ for $p < \infty$; pointwise supremum for $p = \infty$. Then $\epsilon_p = \nu \int |\text{grad } u|^2 = \nu \lambda^{4-d} \int |\text{grad } F|^2$ and $\|u\|_\infty = \lambda \|F\|_\infty$, so

$$K[u] = \lambda \|F\|_\infty / (\lambda^{4-d})^{1/3} / (\nu \int)^{1/3} \\ = \lambda^{(d-1)/3} K[F].$$

No rate or history of λ enters: only its instantaneous value. QED.

Verification: experiments/log_corridor.py constructs snapshots at γ in $\{0, 1/2, 1\}$; all nine fitted slopes equal $(d-1)/3$ to five decimals across four decades of s .

LEMMA 5 (Irrotationality of the self-similar family). Let $u = A(r) e_r$ with A smooth and $A(0)$ finite (radial velocity field). Then $\omega = \text{curl } u = 0$ identically: the family is irrotational.

Proof. $u_x = A(r) x/r$, $u_y = A(r) y/r$. By the chain rule:

$$\begin{aligned} d(u_y)/dx &= A'(r)(x/r)(y/r) + A(r) d(y/r)/dx \\ &= A'xy/r^2 + A(r)(-xy/r^3) \\ &= xy/r^2(A' - A/r). \\ d(u_x)/dy &= A'xy/r^2 + A(r)(-xy/r^3) \\ &= xy/r^2(A' - A/r). \end{aligned}$$

Hence $\omega_z = d(u_y)/dx - d(u_x)/dy = 0$ for all $r > 0$. At $r = 0$: $u_x = u_y = 0$ (removable singularity), $\omega = 0$ by continuity. In 3D: same argument for each curl component. QED.

Corollary. The self-similar concentrating family (type-I: $A(r) = \lambda F(\lambda r)$; type-II: $A(r) = \lambda^{d-1} F(\lambda r)$) is irrotational for all $d \geq 1$ and all $\sigma \geq 1/2$.

Consequence for criteria. enstrophy $= \int |\omega|^2 dx = 0$ and $L^p(\omega) = 0$ for every p , for every concentrating snapshot. These functionals measure exactly zero. Meanwhile $\|u\|_\infty$ grows without bound. The supremum is the unique functional that detects the family.

References

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Companion verification scripts: experiments/outward_cascade.py, outward_cascade_extended.py, selfsimilar_cascade.py, selfsimilar_type2.py; data outputs in data/*.json.